

E-COMMERCE WEBSITE FOR ISABELLE CONCEPTS & PRINTS

A Project

Presented to

The Faculty of the College of Information Technology and Engineering

De La Salle Lipa

In partial fulfillment

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Information Systems Analysis and Design and IT Project Management

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Chapter 1

BUSINESS CASE

Isabelle Concept & Prints, a custom printing business specializing in souvenirs, personalized gifts, and promotional materials, currently relies on Facebook to handle customer inquiries and showcase products. However, this manual approach results in inefficiencies, such as delayed responses, repetitive inquiries, and missed sales opportunities. The absence of an e-commerce platform limits the company's reach and prevents customers from placing orders conveniently.

To address these challenges, the proposed e-commerce website will serve as a centralized platform where customers can browse products, customize orders, and track purchases in real time. This solution will improve customer experience, reduce manual workload, and allow the business to operate 24/7. By automating inquiries through an FAQ section, streamlining order processing, and enhancing order tracking, the website will eliminate inefficiencies and ensure seamless transactions. Furthermore, it will create new revenue opportunities by making the business more accessible to a wider customer base. Ultimately, the implementation of an e-commerce website will enable Isabelle Concept & Prints to enhance operational efficiency, improve customer engagement, and remain competitive in the digital market.

SWOT ANALYSIS OF THE CURRENT SYSTEM

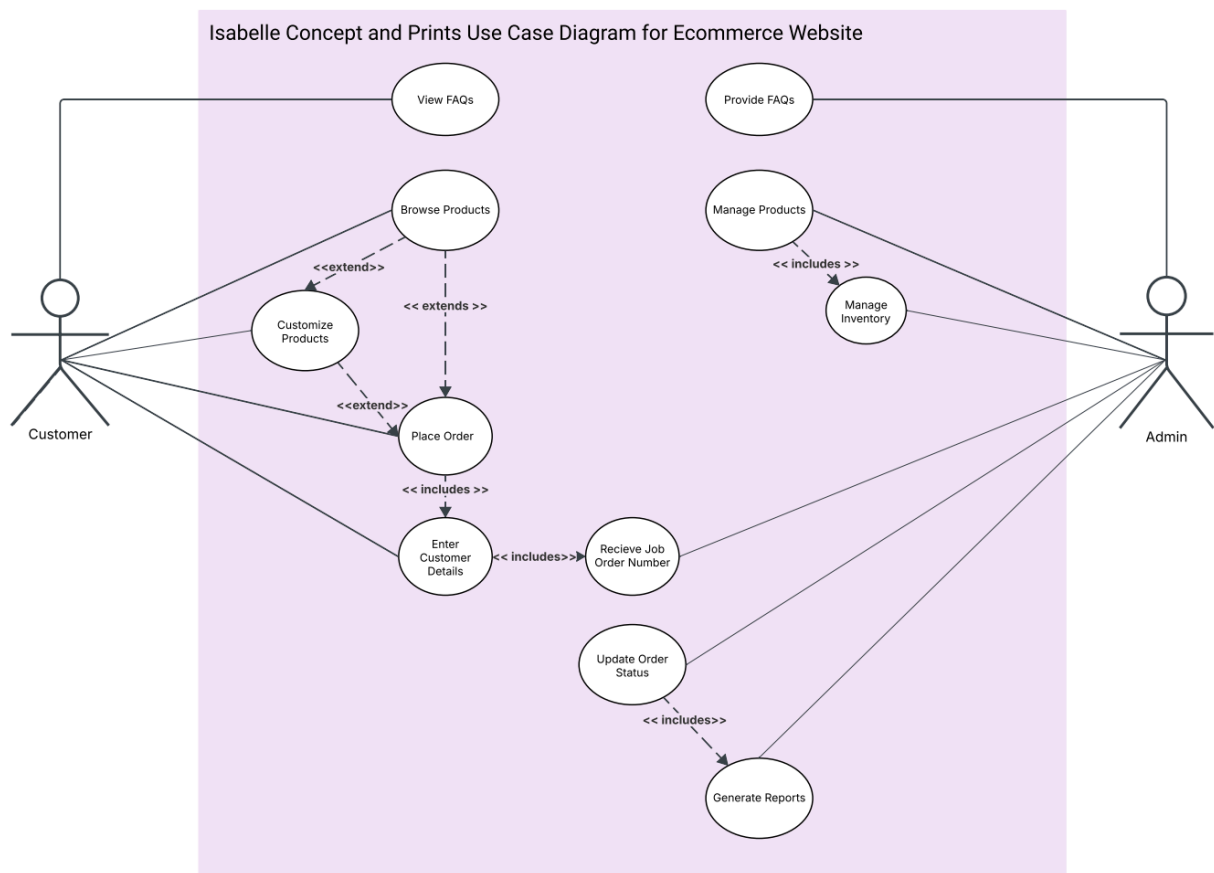
Strengths: The current system, which relies on Facebook for customer interactions and UTAK POS for sales transactions, offers some benefits. It is user-friendly, requiring minimal training for staff to manage inquiries and sales. Additionally, social media presence provides free marketing and customer engagement, making it easy to showcase products and attract customers. The business also benefits from an existing barcode system for inventory tracking, which helps streamline stock monitoring and reduce errors in manual record-keeping.

Weaknesses: Despite its advantages, the current system has significant inefficiencies that hinder business growth. The reliance on manual responses for customer inquiries and order placements leads to delays and lost sales opportunities. The absence of an automated ordering process prevents customers from making purchases independently, limiting business operations to working hours. Furthermore, UTAK POS lacks e-commerce functionality, making it inadequate for handling online transactions. The system also does not provide real-time order tracking, forcing customers to follow up manually, adding to the workload.

Opportunities: By implementing an e-commerce website, Isabelle Concept & Prints can automate customer inquiries, allow 24/7 online ordering, and enhance customer experience. This shift would enable seamless integration of order processing, product customization, and inventory management, reducing manual workload and increasing efficiency. The website can also expand the company's reach, attracting new customers beyond its current social media audience.

Threats: Transitioning to an e-commerce platform comes with potential risks, such as resistance to change from customers accustomed to Facebook inquiries. Cybersecurity threats, including data breaches and unauthorized access, must also be addressed to protect customer and business data. Furthermore, technical challenges during system implementation, such as compatibility with existing processes and data migration issues, could cause disruptions or delays. Lastly, market competition from other online stores with advanced digital capabilities poses a challenge, making it essential for the new system to be well-optimized and customer-friendly.

USE CASE DIAGRAM



The use case diagram illustrates the functional requirements of the proposed e-commerce system for Isabelle Concepts & Prints. The system involves two primary actors and several key interactions:

Customer Actor:

Customers can browse the product catalog, search for specific items, and view detailed product information including pricing and customization options. They can customize products by adding text, selecting colors, or uploading design files. The system allows customers to add items to their shopping cart, place orders, and track order status in real-time. Optional account creation enables faster checkout and order history viewing.

Admin/Staff Actor:

Administrative users can manage the entire product catalog, including adding new products, updating descriptions, and managing inventory levels. They process incoming orders, update order status throughout the production cycle, and generate sales reports for business analysis. Staff can also manage customer accounts and respond to customer inquiries through the system.

The diagram demonstrates the system's core functionality of enabling independent customer ordering while providing comprehensive administrative tools for efficient business management.

FEASIBILITY ASSESSMENT

- a. **Operational Analysis**—Provide an analysis of how the proposed system will work effectively within the organization in terms of the following :

Project Objectives Alignment

The proposed e-commerce website and production management system align closely with Isabelle Concepts & Prints' goals of streamlining operations and expanding digital presence. The system addresses key business challenges such as inefficient order handling, lack of real-time tracking, and limited customer accessibility by providing a centralized platform for order placement, tracking, and communication. Through automation and better workflow integration, the business can operate more efficiently while improving customer satisfaction.

Impact on Organizational Structure & Procedures

The new system will introduce changes in workflow by shifting from manual Facebook-based inquiries to a

semi-automated digital platform. Employees will transition to managing orders and updates via the system rather than responding individually to each customer. Minimal training will be needed, as the system is designed to be user-friendly, focusing on order approval, production status updates, and inventory input. While some initial resistance may occur due to adjustment challenges, this can be mitigated through guided training and hands-on onboarding.

Consistency with Existing Systems & Policies

The project supports the company's informal IT strategy of adopting digital solutions to improve efficiency. While the current POS (UTAK POS) is still in use, it cannot fully meet the operational demands of the business. The proposed system will function independently but is designed to be compatible with the business's existing POS hardware and workflows. Since financial transactions are handled outside the platform, regulatory concerns are limited to data privacy, which will be addressed through role-based access and compliance with data protection standards.

Potential Risks & Challenges

Risks include potential compatibility issues with existing tools, customer adjustment to the new platform, and data migration inconsistencies. These can be mitigated through the following strategies:

Compatibility Testing *before deployment to ensure the system works with existing barcode tools and hardware.*

Structured Data Migration *plan with validation and backups.*

Customer Transition Period *with continued Facebook support and encouragement to shift to the new platform.*

Secure Hosting & Regular Backups *to minimize risks of system downtime or data loss.*

- b. Technical Analysis** - Provide an analysis of whether the proposed system can be implemented using the company's available technology, skills, and infrastructure.

Researchers should ask and analyze the following :

Hardware and Infrastructure Requirements

Isabelle Concepts & Prints does not maintain an in-house IT infrastructure but operates through existing POS hardware and basic networking equipment. Since the proposed system will be cloud-based, no significant upgrades to servers or local infrastructure are needed. Cloud hosting is a viable and cost-efficient alternative, offering reliability, scalability, and ease of access from any device.

System Compatibility and Integration

The system is not designed to integrate directly with UTAK POS but will operate alongside it. As such, there are no expected issues with database compatibility or ERP/CRM integration. However, inventory and job order data from the current system will need to be manually migrated, which will be handled carefully through a structured process to ensure data accuracy and continuity.

Maintenance Requirements

Since the business does not have a dedicated IT team, the system will be developed for ease of use and minimal maintenance. The backend will be simple to navigate, and role-based access will allow authorized staff to manage key functions like status updates, inventory entries, and custom order approvals. Occasional support or updates may require assistance from the development team or external IT consultants.

Cost and Resource Availability

The main costs will involve website development, hosting, and domain registration. To minimize expenses, the system will utilize open-source tools for core functionalities. Any required paid features or third-party tools (such as for optimization or email integration) will be covered by the business. No additional hardware purchases are necessary, making the implementation feasible under a moderate budget. A maintenance budget will also be set aside to ensure the platform remains updated and secure in the long term.

c. Cost – Benefit Analysis

One-Time Costs (Year 0)	
Cost Component	Amount (PHP)
Web Development	12,000

Training & Support	7,000
Total One-Time Cost	19,000

Recurring Costs (Annual Estimates)	
Cost Component	Amount (PHP)
Domain & Hosting Fees	2,500
Website Maintenance	3,000
Content & Support Updates	2,000
Total One-Time Cost	7,500

Benefits (Annual Estimates)	
Benefit Component	Amount (PHP)
Revenue from New Online Orders (estimated 10% gain)	2,500
Savings from Reduced Manual Workload (~20% less labor)	3,000
Extended Market Reach (branding, digital exposure)	2,000
Total One-Time Cost	7,500

Net Benefits Over 5 Years (with 10% discount rate)					
Year	Benefits	Costs	Net Benefits	Discount Factors	Present Value
0	0	19,000	-19,000	1.0	-19,000
1	7,500	7,500	0	0.91	0
2	7,500	7,500	0	0.83	0
3	7,500	7,500	0	0.75	0

4	7,500	7,500	0	0.68	0
5	7,500	7,500	0	0.62	0
Total NPV				-19,000	

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	Total Benefits (Yearly)	7,500

	Year 0						
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Net economic benefit		₱45,000.00	₱45,000.00	₱45,000.00	₱45,000.00	₱45,000.00	
Discount Rate (10%)	₱0.10	₱0.91	₱0.83	₱0.75	₱0.68	₱0.62	
PV of benefits		₱40,909.09	₱37,190.08	₱33,809.17	₱30,735.61	₱27,941.46	
NPV of all BENEFITS		₱40,909.09	₱37,099.17	₱311,908.34	₱142,643.95	₱170,585.40	₱170,585.40
One-Time Costs	-₱19,000.00						
Recurring Costs	₱0.00	-₱7,500.00	-₱7,500.00	-₱7,500.00	-₱7,500.00	-₱7,500.00	
Discount Rate (10%)	₱0.10	₱0.91	₱0.83	₱0.75	₱0.68	₱0.62	
PV of Recruiting Costs		-₱6,818.18	-₱6,198.35	-₱5,634.86	-₱5,122.60	-₱4,666.91	
NPV of All COSTS	-₱19,000.00	-₱25,818.18	-₱32,016.53	-₱37,651.39	-₱42,773.99	-₱47,430.90	-₱47,430.90
Overall NPV							₱123,154.50
OVERALL ROI - (Overall NPV/NPV of All COSTS)							₱2.60
Break-Even Analysis							
Yearly NPV Cash Flow	-₱19,000.00	₱34,090.91	₱30,991.74	₱28,174.31	₱25,613.00	₱23,294.55	
Overall NPV Cash Flow	-₱19,000.00	₱15,090.91	₱46,082.84	₱74,256.95	₱99,869.95	₱123,154.50	
Project Breakeven occurs in between Year 0 to Year 1							
Use first year of positive cash flow to calculate break-even fraction - ((34,090.91 - 15,090.91) / 34,090.91) =	₱0.56						
Project Breakeven occurs at 0.6 years							

Financial Analysis:

- Break-even Point: The project does not break even within the 5-year analysis period based on current projections
- Net Present Value: -PHP 19,000 (negative)

- Strategic Investment: This project represents a strategic investment in digital transformation rather than immediate financial returns

Non-Financial Benefits:

- Enhanced professional image and market credibility
- Improved customer satisfaction through 24/7 accessibility
- Foundation for future digital expansion and advanced features
- Competitive positioning in the local printing market
- Operational efficiency improvements and reduced manual workload
- Scalability for future business growth

Recommendation:

While immediate financial returns are limited, this project should be considered a necessary strategic investment for long-term competitiveness and digital transformation. The non-financial benefits and future growth potential justify the initial investment.

Chapter 2

PROJECT OBJECTIVES

In order to achieve success in the Isabelle Concept & Prints E-Commerce Website Project, the following objectives must be met within the designated timeline and available resources:

- Develop and deploy a user-friendly e-commerce website that enables customers to browse products, place orders, and track purchases within 90 days from project initiation.
- Implement a secure and automated order processing system that reduces manual workload and ensures 24/7 service availability.
- Add a feature that lets customers track their orders in real-time to improve their experience.
- Incorporate an FAQ section and automated inquiry handling system to minimize repetitive customer service tasks.
- Design and implement a role-based access control system to protect sensitive business and customer data in compliance with data privacy standards.
- Conduct training sessions for staff to ensure smooth use of the new system and help them guide customers.

PROJECT DELIVERABLES

The following key deliverables must be completed for the successful delivery of the Isabelle Concept & Prints E-Commerce Website Project:

1. A fully working and easy-to-use e-commerce website where customers can browse, enable basic customization, and order products.
2. Real-time order tracking included in the website.
3. A user manual that shows how to manage orders, update products, and use the system.
4. A simple training session for employees to learn how to use the new website.
5. A secure login and access system to protect user and business information.
6. Backup and recovery plan to keep the system safe from data loss.

PROJECT CONSTRAINTS

1. The new e-commerce website must operate alongside the current UTAK POS and barcode systems used by Isabelle Concept & Prints. Since these tools will not be replaced or modified, the website must be designed to complement existing processes without requiring full system integration. This limits how data is shared between the old and new systems and requires manual migration of inventory and job order data.
2. The business does not maintain in-house IT infrastructure such as servers or advanced networking equipment. As a result, the e-commerce system must be cloud-hosted using reliable third-party services that are affordable, secure, and accessible on any device. This constraint eliminates the option of a locally hosted system and requires careful selection of hosting platforms that fit the company's budget and performance needs.
3. To accommodate the small project team, limited time, and evolving user needs, the project will follow an Agile software development methodology. This includes delivering the system in small, manageable parts (sprints), incorporating feedback from early testing, and adjusting features based on the actual use of the prototype. The project will not follow heavy documentation models like the Rational Unified Process (RUP), due to its complexity and overhead.
4. Isabelle Concept & Prints does not have an in-house IT team. Therefore, the system must be easy to use and maintain by non-technical staff. All administrative functions, such as updating inventory or approving custom orders, must be manageable through a simple interface. Any complex technical tasks (e.g., server maintenance or troubleshooting) will rely on external support, which

may cause delays if not minimized.

5. The project has a limited budget, covering basic website development, domain registration, and hosting. To reduce costs, the system must rely on open-source tools and avoid expensive third-party software. Additional features, such as email integration or premium analytics, must be carefully evaluated for affordability and necessity before inclusion.

PROJECT ASSUMPTIONS

1. Open-source development tools and existing laboratory hardware will be used; no commercial software or paid services are assumed necessary within the project scope.
2. The current hardware (including student laptops and lab PCs) will be sufficient for development and testing; no new hardware purchases are assumed.
3. Internal staff of Isabelle Concepts & Prints will cooperate in testing, feedback, and basic training for the e-commerce platform.
4. The digitization of existing product data (e.g., images, descriptions, prices) will be facilitated by the business and made available to the development team in a timely manner.
5. Hosting will be cloud-based and managed using cost-effective or free-tier platforms without requiring dedicated servers or high-end infrastructure.
6. The student team will be able to complete the project within the academic semester timeline using existing technical skills and guidance from faculty.
7. Stakeholders will be available for scheduled feedback sessions during key development and testing phases.
8. Basic internet access and local testing environments will be sufficient for all development, deployment, and demonstration needs.

PROJECT DESCRIPTION

The Isabelle Concepts & Prints E-Commerce Website Project will cover the following processes related to digital customer engagement and internal production management:

- a. Online product browsing and ordering
- b. Basic order customization and upload handling
- c. Order status tracking and customer notification
- d. Internal job order and inventory tracking
- e. Staff training

Major Functions of the E-Commerce System

Major Functions	Description
Product Catalog and Display	The system will allow customers to browse a categorized catalog of customizable products, each with detailed descriptions, images, and pricing.
Order Placement and Customization	Customers can place orders online with basic customization options such as text input, color selection, and image/file uploads for personalized printing.
Order Tracking and Status Updates	Customers will receive updates on their order status (e.g., processing, in production, ready for pickup/shipping) via the website interface or email.
Job Order and Inventory Tracking	Internal staff can view, update, and manage job orders, track materials used, and monitor inventory levels relevant to each printing task.
User Account Management (Optional)	Customers may optionally create user accounts for faster checkouts, saved preferences, and viewing order history.
Basic Customer Inquiry and Contact Tools	The system will include a basic contact form or inquiry module.

PROJECT TEAM

Project Sponsor

Isabelle Concepts & Prints - Business Owner

Acts as the primary stakeholder and beneficiary of the project. Provides business insights, approves deliverables, and ensures the system aligns with operational needs.

Project Manager

Regina Angeli Cadeliña - Project Manager / Full Stack Developer

Oversees the overall execution of the project, manages the timeline, coordinates team tasks, and leads the front-end design, back end integration and development of the website interface.

IT Team

Juan Benjo Estrella - Lead Developer / Full Stack Developer

Responsible for backend and front-end development, database integration, system functionality, and technical problem-solving.

Janna Mikhaela Alcantara - System Analyst

Jeric Lique - System Analyst

Gathers and analyzes business requirements, prepares technical documentation, and supports the team in aligning the system to user needs.

RACI MATRIX

RACI Table for E-Commerce Website Development Project (Agile Methodology)				
SDLC Phase & Task	Project Manager (Regina)	Lead Developer (Benjo)	System Analysts (Janna & Jeric)	Business Owner
1. Requirements Gathering & Analysis				
1.1 Define System Requirements	R, A	C	R	C
1.2 Conduct Feasibility Study	R, A	C	R	I
1.3 Stakeholder Alignment	R, A	I	C	C
2. System Design				
2.1 Design System Architecture	C	R, A	C	I
2.2 Design User Interface (UI/UX)	R, A	C	C	C
2.3 Design Database Schema	C	R, A	C	I
3. Development				

3.1 Setup Development Environment	A	R	I	I
3.2 Front development	R, A	C	I	I
3.3 Backend Development	C	R, A	I	I
3.4 Database Implementation	C	R, A	I	I
4. Testing				
4.1 Unit Testing	C	R, A	I	I
4.2 Integration testing	R, A	R	C	I
4.3 User Acceptance Testing	C	R, A	I	I
5. Deployment				
5.1 Prepare for Launch	C	R, A	I	I
5.2 Execute System Launch	R, A	R	I	A
5.3 Monitor Initial Performance	R, A	R	I	C
6. Maintenance & Support				
6.1 Provide Ongoing Support	A	R	C	I
6.2 Implement Updates	A	R	C	C
6.3 Staff Training	R, A	C	C	R

Role Explanations:

Project Manager (Regina Cadelíña):

- Accountable for overall project success and timeline management
- Responsible for UI/UX design and frontend and backend development
- Coordinates all team activities and stakeholder communication

IT Team

Lead Developer (Juan Benjo Estrella):

- Responsible for backend and front end development and technical architecture
- Accountable for system functionality and technical implementation
- Provides technical consultation for design decisions

System Analysts (Janna Alcantara & Jeric Lique):

- Responsible for requirements gathering and documentation
- Consulted on design and testing phases
- Support project coordination and user needs analysis

Business Owner (Isabelle Concepts & Prints):

- Accountable for business requirements and final acceptance
- Consulted on major design and functionality decisions
- Responsible for user acceptance testing and staff training participation

WORK BREAKDOWN STRUCTURE

The Work Breakdown Structure (WBS) is essential for this e-commerce project as it breaks down the complex development process into manageable, measurable tasks. It enables effective resource allocation, timeline management, and progress tracking throughout the Agile development cycle. The WBS ensures that all project deliverables are identified and properly sequenced, facilitating better coordination among team members and stakeholders.

WBS for E-Commerce Website Development Project (Agile Methodology)					
WBS Code	Task/Activity	Resources (Team, Hardware & Software)	Dependencies	Effort (Hours)	Duration (Weeks)
1	Sprint 0: Project Initiation & Planning	Project Team, Laptops, Documentation Tools	none	80	4
1.1	Define Project Scope	All Team Members	none	20	1
1.2	Team Onboarding & Roles	All Team Members	1.2	20	1
1.3	Setup Dev Tools & Repository	Lead Developer, Project Manager	1.3	20	1
1.4	Requirement Gathering	System Analysts, Business Owner	1.3	20	1
2	Sprint 1: Basic Website	Frontend Team, Design	1	80	4

	& Landing Pages	Tools, Web Server			
2.1	Build Home Page	Project Manager, Lead Developer	1.4	40	2
2.2	About, Services, and FAQ Pages	All Team Members	2.1	20	1
2.3	Business Info & Contact Page	All Team Members	2.2	20	1
3	Sprint 2 : Product Catalog & User System	Full Team, Database Server, Testing Tools	2	80	4
3.1	Create Product Listing Page	Lead Developer, Project Manager	2.3	40	2
3.2	Build Product Listing Page	Lead Developer, Project Manager	3.1	20	1
3.3	Implement User Login & Registration	Lead Developer	3.2	20	1
4	Sprint 3 : Custom Order & Cart Features	Full Team, Payment Testing Environment	3	80	4
4.1	Implement User Login & Registration	Lead Developer	3.2	20	1
4.2	Add Customization Form & Preview Logic	Lead Developer, Project Manager	3.3	40	2
4.3	Order Summary + Customer Confirmation	Lead Developer, Project Manager	4.2	20	1
5	Sprint 4: Order Tracking & Admin Panel	Full Team, Admin Interface Tools	4.0	80	4
5.1	Develop Order Management Dashboard	Lead Developer, System Analysts	4.2	40	2
5.2	Status Update Functionality	Lead Developer	5.1	20	1
5.3	Inventory Monitoring (basic integration)	Lead Developer, System Analysts	5.2	20	1
6	Sprint 5: Testing,	Full Team, Production	5	80	4

	Review & Deployment	Server, Training Materials			
6.1	User Testing & Bug Fixes	All Team Members	5.3	20	1
6.2	Feedback Iteration & Final Polishing	All Team Members	6.1	20	1
6.3	Deployment & Go-Live	Lead Developer, Project Manager	6.2	20	1
6.4	Staff Training & Handover	All Team Members	6.3	20	1

Total Project Duration: 20 Weeks

Total Effort: 400 Hours

MILESTONES

The table below lists the milestones for this project, along with their estimated completion timeframe.

Milestone	Target Date	Deliverables
Project Initiation Complete	Week 4	- Project scope, team roles, and development environment established
Basic Website Launch	Week 8	- Functional website with home page, about, services, and contact pages
Product Catalog Implementation	Week 12	- Complete product listing and detail pages with user registration system
E-commerce Functionality	Week 16	- Shopping cart, customization features, and order confirmation system
Admin Panel & Order Tracking	Week 20	- Order management dashboard and inventory monitoring system
System Deployment & Training	Week 24	- Live website deployment and staff training completion

Risk Management.

Risk Management, as defined by the AS/NZS 4360:2004 standard, is “the systematic application of management policies, procedures, and practices to the tasks of establishing the context, identifying, analyzing, assessing, treating, monitoring, and communicating risks.” It is a structured approach that allows organizations to understand and control the uncertainty that may affect their goals and outcomes.

This process is not a one-time activity but an ongoing, iterative cycle that evolves alongside the project or organization. With each iteration, the organization gains a deeper understanding of the nature and severity of potential risks, enabling more informed decision-making. Through continuous monitoring and review, risk management contributes to better resource allocation, increased resilience, and enhanced operational performance.

Risk, in this context, is broadly defined as “the possibility of an event occurring that may impact the achievement of objectives.” These risks can stem from various sources—including people, processes, technology, and data—and may manifest as delays, cost overruns, technical failures, or external disruptions. By identifying these possibilities early, organizations can proactively develop mitigation strategies and contingency plans to reduce the likelihood of negative outcomes or lessen their impact.

Ultimately, effective risk management ensures that projects like the Isabelle Concepts & Prints E-Commerce Website are delivered on time, within budget, and according to quality standards—despite the uncertainties that may arise throughout the development lifecycle.



Figure 1. Risk Management Process

The risk management process involves continuous steps:

1. Risk Identification
2. Risk Analysis
3. Risk Prioritization
4. Risk Response Planning
5. Risk Mitigation
6. Risk Monitoring
7. Risk Documentation and Review

1. Risk Identification

Potential risks have been categorized based on the People, Process, Technology, and Data dimensions and aligned with key milestones of the project.

Project Phase	Milestone	Risk Factor	Potential Failure	Effects of Potential Failure
Initiation & Planning	Project Initiation Complete	Process	Unclear or shifting project scope	Misaligned expectations, feature creep
		People	Unclear team responsibilities	Task overlap or missed deadlines
		Technology	Incompatible tools or repository setup issues	Development delays and sync issues
		People	Stakeholder availability issues	Incomplete requirements, needing rework later
Basic Website Build	Basic Website Launch	Technology	Browser/device compatibility	Broken layout, poor user experience
		People	Missing or unclear content	Poor information delivery to

				customers
		Process	No email validation or contact form issues	Missed customer inquiries
Catalog & Cart Features	Product Catalog Implementation	Process	Poor product categorization	Confusing user navigation
		Security	Weak password policy/no account validation	User data vulnerability, unauthorized access
Order & Cart Features	E-commerce Functionality	Technology	File upload or preview function fails	Incomplete orders or user frustration
		Process	Calculation or display errors	Wrong order totals or confirmation confusion
Order Tracking/Admin	Admin Panel & Order Tracking	Technology	Dashboard doesn't reflect real-time data	Delayed order processing and inventory mismatch
		Data	Inventory updates not syncing	Stockouts or overselling
Testing & Deployment	System Deployment & Training	Process	Incomplete test cases	Undiscovered bugs in production
		Technology	Server or domain configuration failure	Failed launch or inaccessible site
		People	Inadequate staff knowledge on system usage	Errors in order processing and customer support

2. Risk Analysis

Determining the Probability of Occurrence and the Severity of Impact

Severity of Impact	
0 - Not Applicable	Not Applicable
1 - Minor	Less than 1-day slip in schedule
2 - Serious	Greater than 1-day slip in schedule, 5% cost
3 - Major	Greater than 2-days slip in schedule, 7% cost
4 - Critical	Greater than 3-days slip in schedule, 10% cost

Occurrence	
0	Not Applicable
1	Seldom
2	Occasional
3	Likely
4	Frequent

Project Phase	Milestone	Risk Factor	Potential Failure	Effects of Potential Failure	Severity Rating	Cause of Potential Failure	Occurrence Rating
Initiation & Planning	Project Initiation Complete	Process	Unclear or shifting project scope	Misaligned expectation, feature creep	2	Vague or evolving stakeholder input	3
		People	Unclear team responsibilities	Task overlap or missed deadline	2	No formal task assignments or documentation	2
		Technol	Incompat	Developme	2	Tools not	2

		ogy	ible tools or repository setup issues	nt delays and sync issues		tested before integration	
		People	Stakeholder availability issues	Incomplete requirements, needing rework later	2	Schedule conflicts with business owner	3
Basic Website Build	Basic Website Launch	Technology	Browser/device compatibility issues	Broken layout, poor user experience	2	No cross-browser testing	2
		People	Missing or unclear content	Poor information delivery to customers	2	Lack of finalized content	2
		Process	No email validation or contact form issues	Missed customer inquiries	2	Form not tested with real inputs	2
Catalog & User System	Product Catalog Implementation	Process	Poor product categorization	Confusing user navigation	2	No clear product taxonomy	2
		Security	Weak password policy/ no validation	User data vulnerability, unauthorized access	3	No encryption or validation in place	3
Order & Cart Features	E-commerce Functionality	Technology	File upload or preview function fails	Incomplete orders or user frustration	2	Missing upload validation or logic	3

		Process	Calculation or display errors	Wrong order totals or confirmation confusion	2	Incorrect logic for order total or formatting	2
Order Tracking/Admin	Admin Panel & Order Tracking	Technology	Dashboard doesn't reflect real-time data	Delayed order processing and inventory mismatch	3	Poor data sync with backend	2
		Data	Inventory updates not syncing	Stockouts or overselling	3	No live inventory system or update delays	2
Testing & Deployment	System Deployment & Training	Process	Incomplete test cases	Undiscovered bugs in production	2	No formal test plan or QA process	3
		Technology	Server or domain configuration failure	Failed launch or inaccessible site	3	Incomplete deployment checklist	2
		People	Inadequate staff knowledge on system usage	Errors in order processing and customer support	2	No training materials or walkthroughs provided	3

3. Risk Prioritization

Determining the Risk Rating

Risk Rating = Probability of Occurrence * Severity of Impact

Risk Rating	Risk Priority
1 to 4	1 - Low
5 to 8	2 - Moderate
9 to 12	3 - High
13 to 16	4 - Showstopper

Project Phase	Milestone	Risk Factor	Potential Failure	Severity	Occurrence	Risk Rating	Risk Priority
Initiation & Planning	Project Initiation Complete	Process	Unclear or shifting project scope	2	3	6	Moderate
		People	Unclear team responsibilities	2	2	4	Low
		Technology	Incompatible dev tools	2	2	4	Low
		People	Stakeholder scheduling conflicts	2	3	6	Moderate
Basic Website Build	Basic Website Launch	Technology	Browser/device compatibility issues	2	2	4	Low
		People	Missing or unclear	2	2	4	Low

			content				
		Process	Faulty contact form	2	2	4	Low
Catalog & User System	Product Catalog Implementation	Process	Poor product categorization	2	2	4	Low
		Security	Weak password/account validation	3	3	9	High
Order & Cart Features	E-commerce Functionality	Technology	File upload or preview fails	2	3	6	Moderate
		Process	Order total or confirmation error	2	2	4	Low
Order Tracking/Admin	Admin Panel & Order Tracking	Technology	Dashboard not syncing real-time	3	2	6	Moderate
		Data	Inventory not updating	3	2	6	Moderate
Testing & Deployment	System Deployment & Training	Process	Incomplete test coverage	2	3	6	Moderate
		Technology	Server/domain misconfiguration	3	2	6	Moderate
		People	Lack of staff training	2	3	6	Moderate

- High Priority: 1 Risk
- Moderate Priority: 7 Risks
- Low Priority: 7 Risks

4. Risk Response Planning

This section outlines how each identified risk will be evaluated for its acceptability based on its severity and probability. **Risks with a rating of 6 or below** are considered **acceptable** and can be managed with standard monitoring. Risks with a rating of 6 or below are considered acceptable and can be managed with standard monitoring. Risks with a **rating above 6** are classified as **not acceptable** and require immediate attention through mitigation strategies. This process ensures that the most critical risks are addressed proactively to prevent major project disruptions.

Project Phase	Milestone	Risk Factor	Potential Failure	Severity	Occurrence	Risk Rating	Risk Priority	Risk Acceptability
Initiation & Planning	Project Initiation Complete	Process	Unclear or shifting project scope	2	3	6	Moderate	Acceptable
		People	Unclear team responsibilities	2	2	4	Low	Acceptable
		Technology	Incompatible dev tools	2	2	4	Low	Acceptable
		People	Stakeholder scheduling conflicts	2	3	6	Moderate	Acceptable
Basic Website Build	Basic Website Launch	Technology	Browser/device compatib	2	2	4	Low	Acceptable

			ility issues					
		People	Missing or unclear content	2	2	4	Low	Acceptable
		Processes	Faulty contact form	2	2	4	Low	Acceptable
Catalog & User System	Product Catalog Implementation	Processes	Poor product categorization	2	2	4	Low	Acceptable
		Security	Weak password/account validation	3	3	9	High	Not Acceptable
Order & Cart Features	E-commerce Functionality	Technology	File upload or preview fails	2	3	6	Moderate	Acceptable
		Processes	Order total or confirmation error	2	2	4	Low	Acceptable
Order Tracking/ Admin	Admin Panel & Order Tracking	Technology	Dashboard not syncing real-time	3	2	6	Moderate	Acceptable
		Data	Inventory not updating	3	2	6	Moderate	Acceptable
Testing & Deployment	System Deployment & Training	Processes	Incomplete test coverage	2	3	6	Moderate	Acceptable
		Technology	Server/d	3	2	6	Moderate	Acceptable

		ogy	omain misconfig uration					
		People	Lack of staff training	2	3	6	Moderate	Acceptable

5. Risk Mitigation

This table summarizes mitigation plans for each of the risks that have been rated Not Acceptable or Moderate Priority. Each of these plans is intended to decrease the risk's chance or effect and is allocated to a specific team member with a target due date.

Milestone	Risk Factor	Potential Failure	Severity	Cause of Potential Failure	Occurrence	Risk Rating	Risk Acceptability	Preventive Action
Product Catalog Implementation	Security	Weak password/account validation	3	No encryption or validation logic	3	9	Not Acceptable	Implement password encryption, email verification, and CAPTCHA in registration
Project Initiation Complete	Process	Shifting or unclear project scope	2	Lack of finalized scope documents	3	6	Acceptable	Conduct formal scope validation meetings with stakeholders and lock scope for sprint
Project Initiation Complete	People	Stakeholder scheduling conflicts	2	Limited availability of business owner	3	6	Acceptable	Schedule all stakeholder sessions in

								advance and send follow-ups
E-commerce Functionality	Technology	File upload or preview function fails	2	Missing preview logic or upload limits	3	6	Acceptable	Add file validation, fallback preview logic, and limit upload size
Admin Panel & Order Tracking	Technology	Real-time data not reflected in dashboard	3	No live data sync between DB and UI	2	6	Acceptable	Enable AJAX-based real-time updates or auto-refresh sync from backend
Inventory Monitoring	Data	Inventory not updating properly	3	Manual or outdated data sync	2	6	Acceptable	Use automatic syncing, real-time stock deduction, and inventory update alerts
System Deployment & Training	Process	Incomplete test cases	2	Lack of structured QA checklist	3	6	Acceptable	Develop a comprehensive QA test plan and assign dedicated QA lead
System Deployment & Training	Technology	Hosting/server misconfiguration	3	No pre-deployment check or backup	2	6	Acceptable	Conduct mock deployment, use server checklist, set rollback/backup points

System Deployment & Training	People	Inadequate staff training	2	No training plan or handover materials	3	6	Acceptable	Create visual guides, training manuals, and run live walkthrough sessions
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- 1 High-risk item mitigated with urgent security controls
- 8 Moderate risks mitigated through preventive QA, planning, or technical strategies

Chapter 4

QUALITY MANAGEMENT

This Quality Management Plan outlines the approach to ensure high-quality delivery of the E-Commerce Website for Isabelle Concepts & Prints. It includes all quality assurance (QA) and quality control (QC) activities, defines the project methodology, testing processes, document reviews, responsibilities, timelines, and deliverables.

Project Software Development Methodology

The project will follow the **Agile Methodology** with Scrum framework to ensure iterative development and continuous quality improvement. Below are the phases and their QA/QC activities:

- Sprint Planning and Initiation:**
 - o Define sprint goals and acceptance criteria
 - o Conduct requirement reviews to identify gaps or inconsistencies
 - o Establish quality standards for each deliverable
- Sprint Development:**
 - o Implement features following coding standards and best practices
 - o Perform continuous code reviews and pair programming
 - o Execute unit testing for each developed component
- Sprint Testing:**
 - o Conduct integration testing to ensure component compatibility
 - o Perform functional testing against defined acceptance criteria
 - o Execute usability testing with stakeholder feedback.

4. Sprint Review and Retrospective:

- o Demonstrate completed features to stakeholders
- o Gather feedback for quality improvements
- o Identify process improvements for subsequent sprints

5. Release Preparation:

- o Conduct comprehensive system testing
- o Perform user acceptance testing (UAT) with business stakeholders
- o Execute deployment testing in staging environment

6. Post-Release Support:

- o Monitor system performance and user feedback
- o Address defects and implement quality improvements

Types of Tests Performed

Test Type	Description	Duration	Responsible
Unit Testing	Verifies individual components or modules for correctness.	Ongoing during development	Lead Developer (Benjo)
Integration Testing	Ensures that modules or components interact correctly.	2-3 days per sprint	Full Development Team
Functional Testing	Tests the entire system's functionality as per requirements.	3-4 days per sprint	System Analysts (Janna & Jeric)
Usability Testing	Evaluates user experience and interface design.	2 days per sprint	Project Manager (Regina) + Stakeholders
Performance Testing	Assesses system response time and load handling capabilities	2 days (Sprint 4)	Lead Developer (Juan Benjo)
Security Testing	Identifies vulnerabilities or potential breaches, especially for IoT applications.	3 days (Sprint 2 & 4)	Lead Developer + External Consultant
Compatibility Testing	Verifies system works across different browsers and device	2 days (Sprint 5)	Full Development Team
User Acceptance Testing (UAT)	Validates the system from the end-user perspective to ensure it meets requirements.	1 week (Sprint 5)	Business Owner + Staff

Document Reviews

Document	Review Type	Frequency	Responsible
Requirement Documents	Peer Review, Stakeholder Validation	Initial + Major Updates	System Analysts, Project Manager
Design Documents	Technical Review, Architecture Walkthrough	Per Sprint	Lead Developer, Project Manager
User Interface Mockups	Design Review, Usability Assessment	Per Sprint	Lead Developer, Project Manager
Code Documentation	Code Review, Technical Documentation Review	Ongoing	Lead Developer, System Analysts
Test Plans and Cases	QA Review, Coverage Analysis	Pre-testing Phase	Project Manager, System Analysts
User Manuals	Content Review, Accuracy Validation	Pre-deployment	Project Manager, System Analysts
Deployment Guides	Technical Review, Process Validation	Pre-deployment	Lead Developer, Project Manager

Quality Responsibilities

Project Manager (Regina Cadeliña):

- Overall quality management oversight
- Usability testing coordination
- Stakeholder communication for quality issues
- Quality metrics reporting

Lead Developer (Juan Benjo Estrella):

- Code quality standards enforcement
- Unit and integration testing execution
- Performance and security testing
- Technical quality reviews

System Analysts (Janna Alcantara & Jeric Lique):

- Requirements quality assurance
- Test case development and execution
- Functional testing coordination
- Documentation quality reviews

QA/QC Activities in Work Breakdown Structure (WBS)

Task ID	Activity	Deliverable	Timeline	Owner
1.1	Requirements Review	Approved Requirements Document	Week 4	System Analysts, Project Manager
1.2	Design Standards Review	Design Guidelines Document	Week 5	Project Manager, Lead Developer
2.1	Code Review Process Setup	Code Review Checklist	Week 6	Lead Developer
2.2	Unit Testing Framework	Unit Test Suite	Week 7	Lead Developer
3.1	Functional Testing	Functional Test Report	Week 10	System Analysts
3.2	Security Testing	Security Assessment Report	Week 11	Lead Developer, External Consultant
4.1	Integration Testing	Integration Test Report	Week 14	Full Development Team
4.2	Performance Testing	Performance Test Report	Week 15	Lead Developer
5.1	Usability Testing	Usability Test Report	Week 18	Project Manager, Stakeholders
5.2	User Acceptance Testing	UAT Report and Sign-off	Week 22	Business Owner, Staff
6.1	Deployment Testing	Deployment Verification Report	Week 23	Lead Developer, Project Manager
6.2	Post-deployment Review	Quality Assessment Report	Week 24	Project Manager